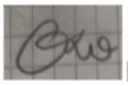
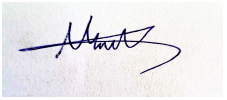
EE3014 Energy Conversion

Experiment 10

THE NO-LOAD OPERATION OF THE TRI-PHASE SYNCHRONOUS MACHINE AS ASYNCHRONOUS GENERATOR

Student No	Name	Surname	Contribution	Signature
150722923	Jacques	Asinyo	Abstract, Introduction, theoretical background	
150725824	Kadir Baha	Demirhan	Results, Conclusion	
150720062	Mehmet Ali	Kamalak	Procedure and Questions	
150722066	Mevlüt	Akhan	Experimental Setup, Procedure and Graphs	

1. Objective

The objective of this experiment is to analyze the performance of a synchronous generator under no-load conditions. Specifically, the experiment aims to:

- Investigate the relationship between the Excitation Current (Ie) and the induced Terminal Voltage (U) while maintaining a constant speed.
- Obtain the Open Circuit Characteristic (OCC) curve, also known as the magnetization curve.
- Observe the effects of magnetic saturation on the generated voltage.

2. Equipment Used

- **Power Supply:** Experiment Board with Energy Unit (Y-036/001).
- **Prime Mover:** Tri-phase Asynchronous Motor (Y-036/015) controlled by an AC Motor Driver (Y-036/026).
- **Generator:** Tri-phase Synchronous Machine (Y-036/021).
- **Measurement:**
    - Energy Analyzer (Y-036/004) for measuring AC output voltage and frequency
    - DC Measurement Unit for measuring Excitation Current (Ie).
    - Tachometer for measuring rotor speed (n).

3. Theory

3.1 Operating Principle

A synchronous generator operates on the principle of electromagnetic induction. A DC current (excitation current, Ie) is fed into the rotor windings, creating a constant magnetic field. The rotor is mechanically driven by a prime mover (in this case, an asynchronous motor) at a synchronous speed (ns).

As the rotor spins, its magnetic field cuts across the stationary stator conductors, inducing an alternating voltage in the stator windings.

3.2 Induced Voltage Formula

The magnitude of the induced EMF (E) per phase in the stator is given by the formula:

$$E = 4.44 * f * N * Phi * Kw$$

Where:

- **E** = Induced Electromotive Force (Voltage)
- **f** = Frequency of the induced voltage

- **N** = Number of turns per phase
- **Phi** = Magnetic flux per pole
- **Kw** = Winding factor

Since the speed is kept constant, the frequency (f) remains constant. Therefore, the induced voltage (E) is directly proportional to the magnetic flux (Phi).

3.3 Open Circuit Characteristic (OCC)

The relationship between the Excitation Current (Ie) and the Output Voltage (U) is not linear forever.

1. **Linear Region:** At low excitation currents, the iron core has low reluctance. The flux increases linearly with the current. This is represented by the "Air Gap Line" in the graph.
2. **Saturation Region:** As excitation increases, the iron core begins to saturate. It requires a much larger increase in current to produce a small increase in flux. The curve bends and flattens out (the "knee" of the curve).

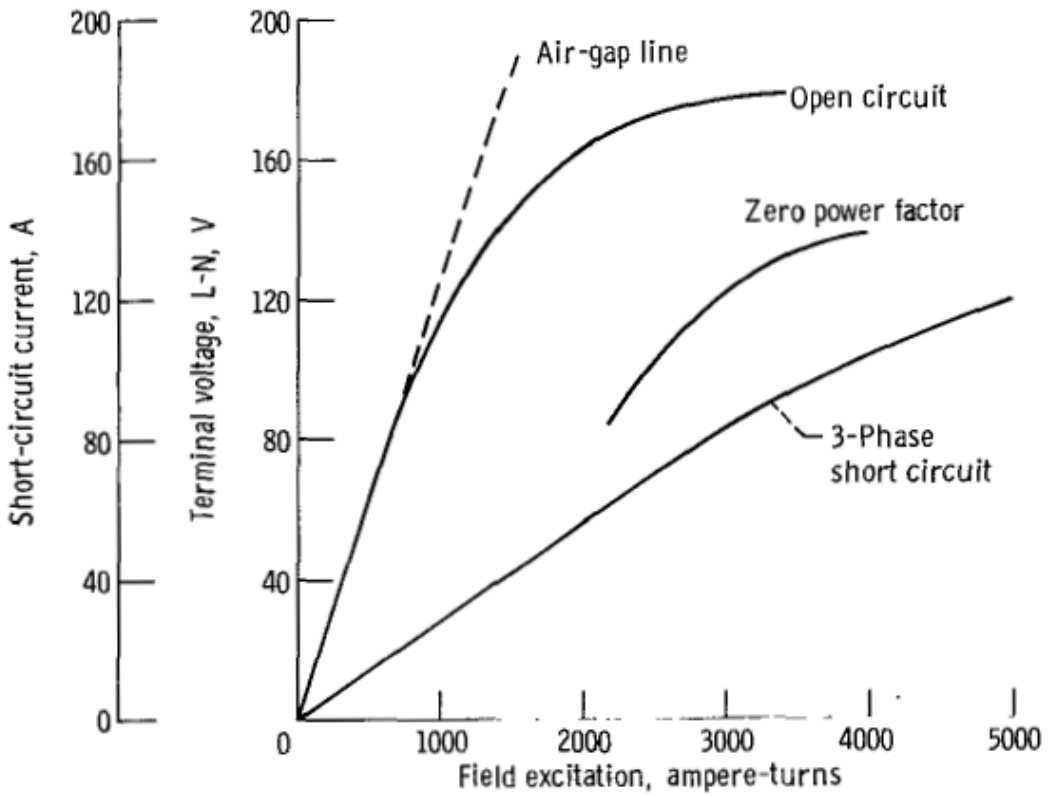
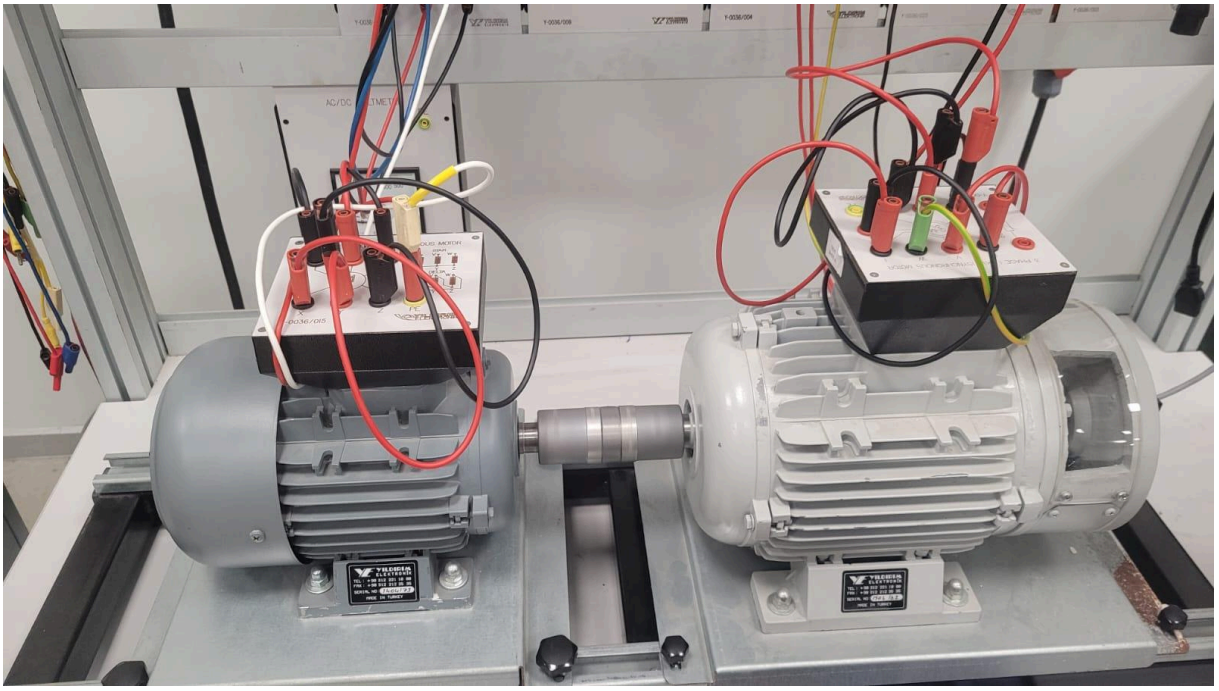


Fig1: Open Circuit Characteristics Curve of Synchronous Generator [1]2

4. Procedure



1. **Connection:** The circuit was connected as shown in the lab manual, coupling the synchronous machine to the asynchronous motor
2. **Startup:** The asynchronous motor was started and brought to the nominal speed of the generator (approx. 1500 rpm).
3. **Measurement:** The excitation current (Ie) supplied to the synchronous generator was increased in steps starting from a low value.
4. **Recording:** At each step, the terminal voltage (U) and Excitation Current (Ie) were recorded while attempting to maintain constant speed.

Connection diagram for the experiment : Y-036/001

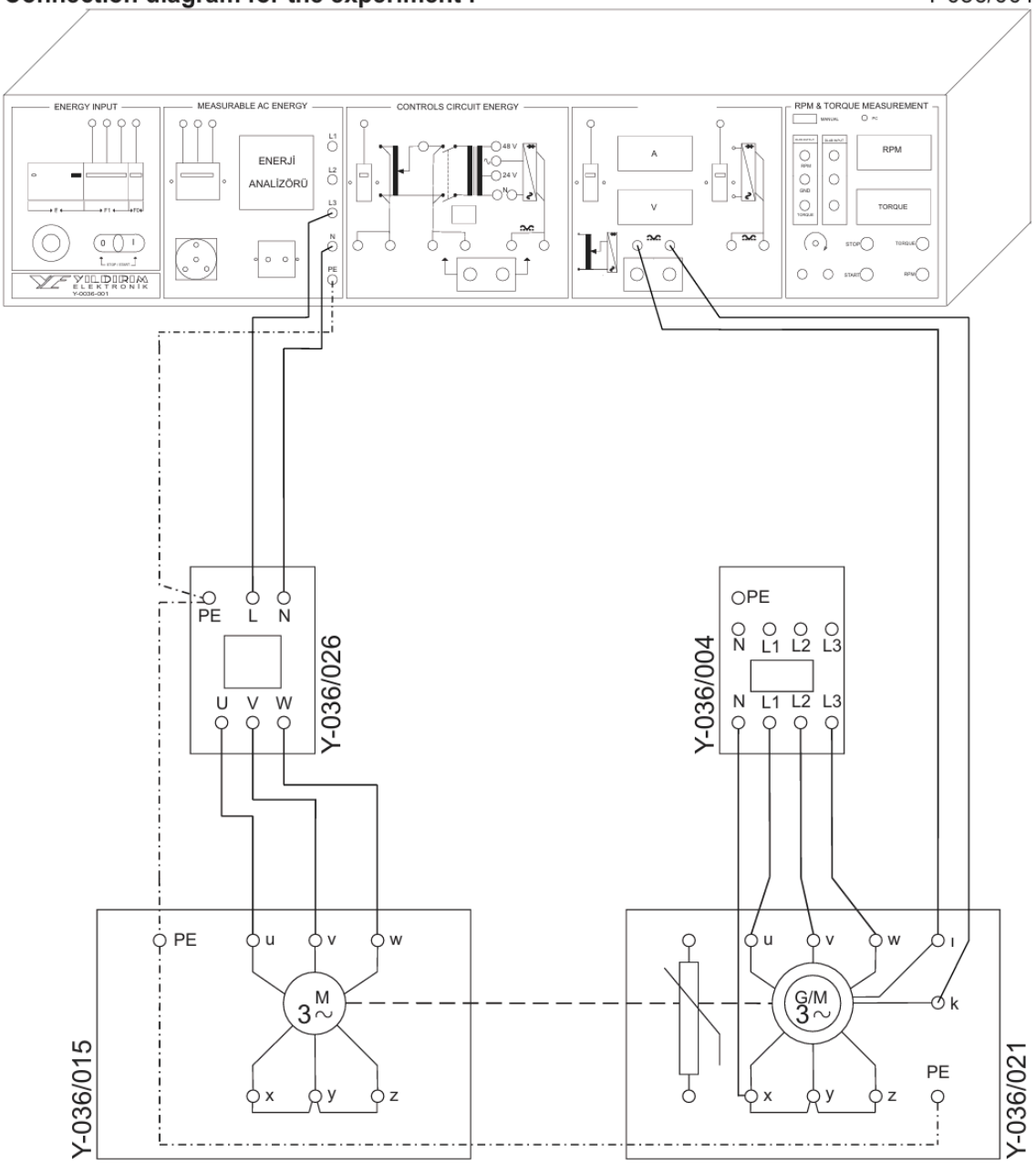


Figure 54.1:Connection diagram for the experiment of tri-phase synchronous generator

Fig2: Connection diagram for the experiment of tri-phase synchronous generator (part1)

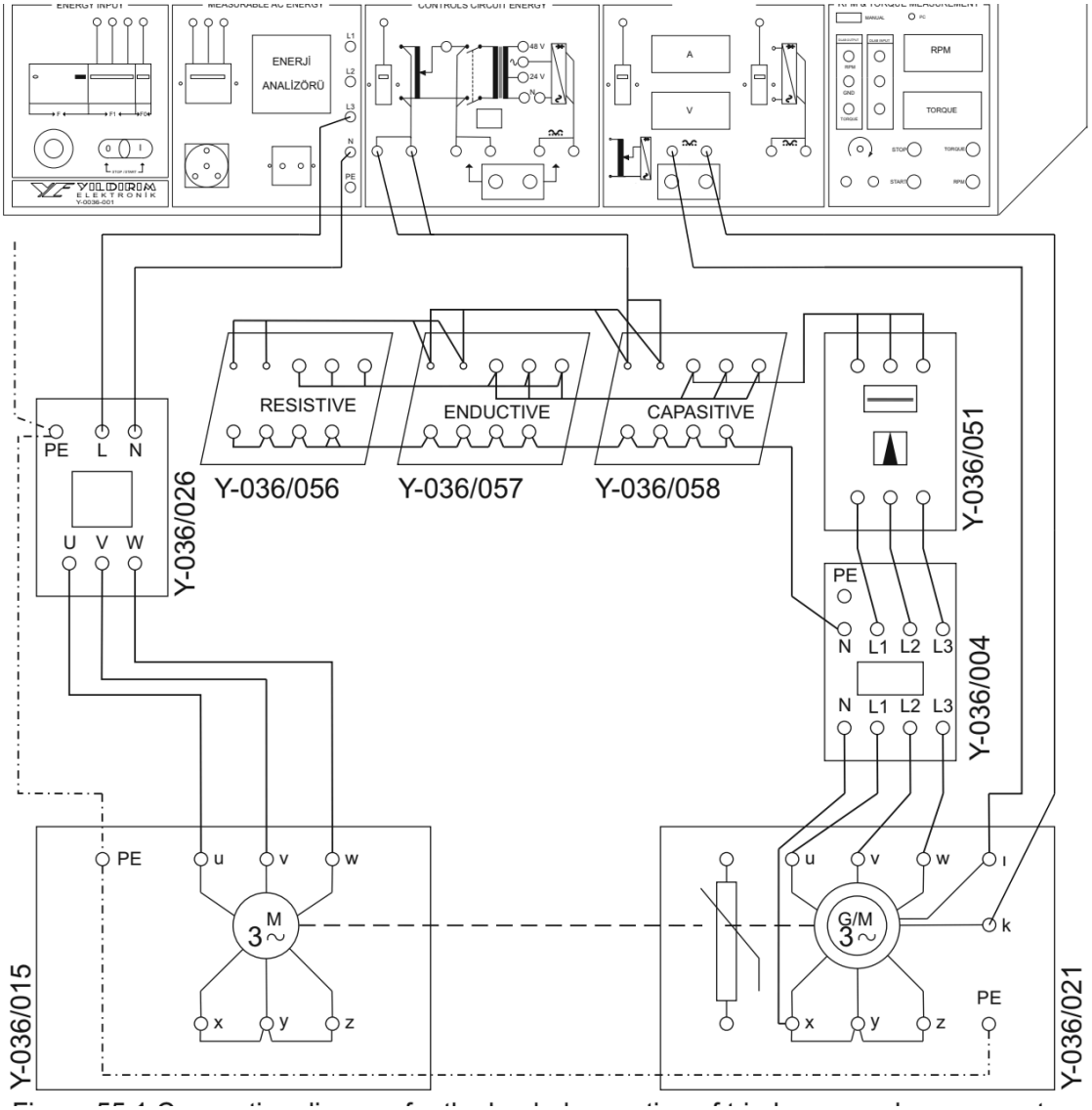
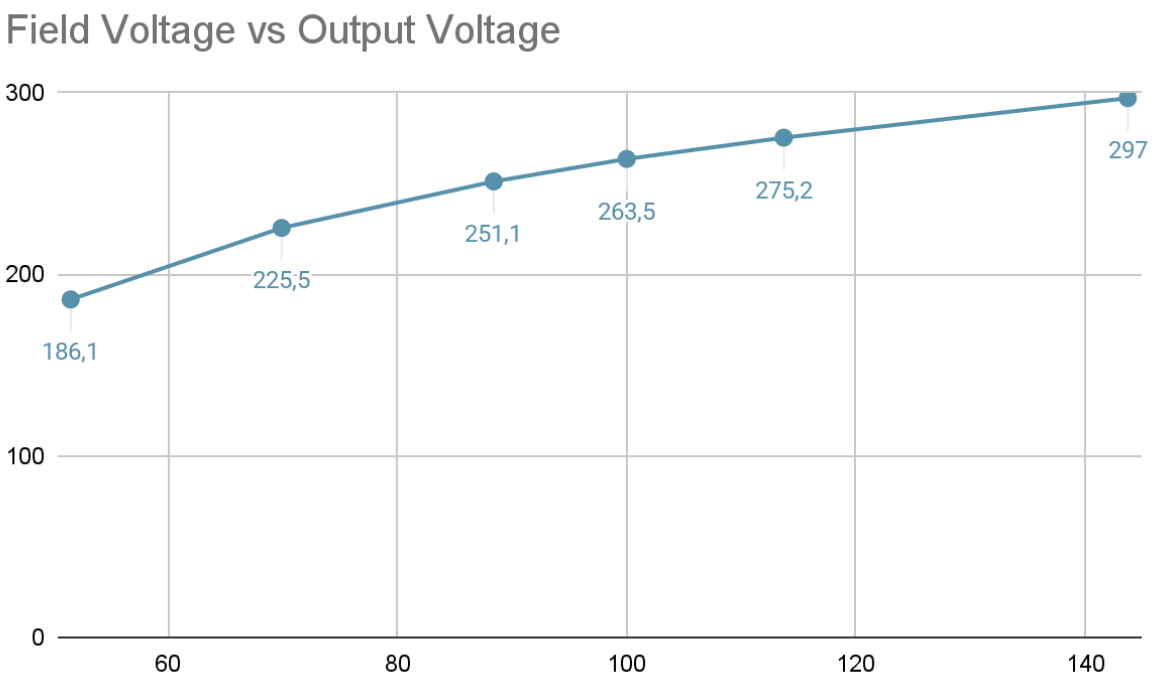


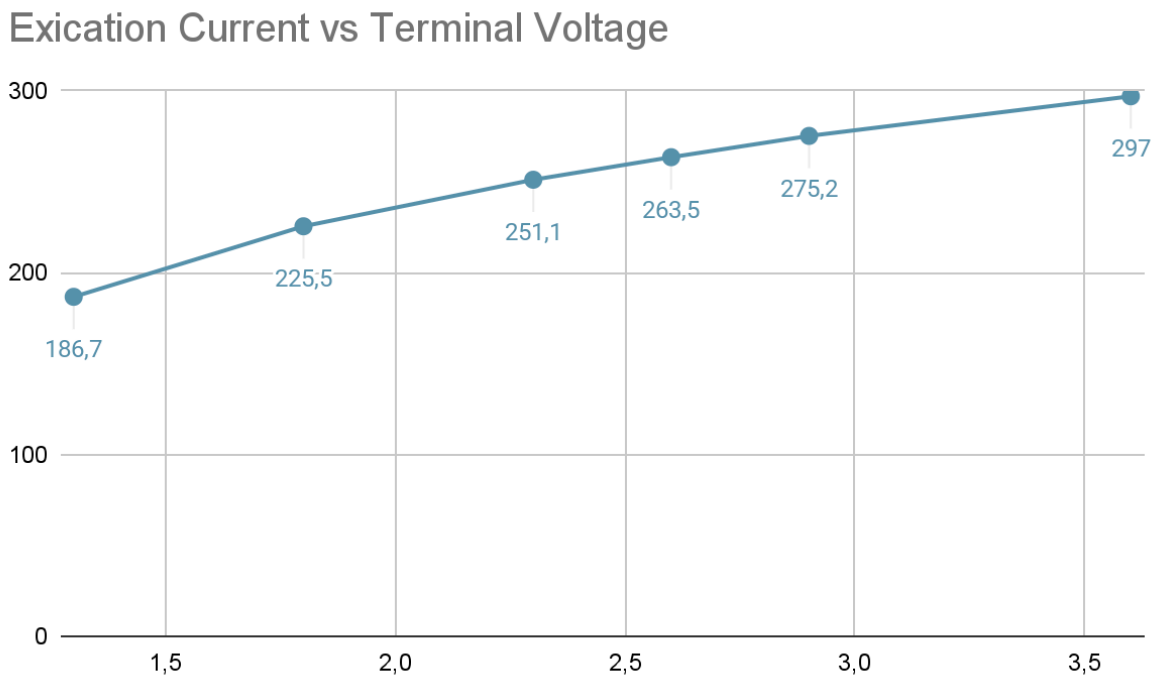
Fig3:: Connection diagram for the loaded operation of tri-phase synchronous motor (part 2)

5. Data and Observations

The following values were recorded during the experiment

Measurement #	Excitation Current (Ie) [A]	Output Voltage (U) [V]	Speed (n) [rpm]
1	1.3	186.7	1500 (approx)
2	1.8	225.5	1500
3	2.3	251.1	1500
4	2.6	263.5	1500
5	2.9	275.2	1500
6	3.6	297.0	1437*





6. Analysis and Discussion

6.1 Analysis of the Magnetization Curve

By plotting the Excitation Current (X-axis) against the Output Voltage (Y-axis), we can observe the magnetic behavior of the generator.

- **Between 1.3A and 1.8A:** The voltage rises sharply (from 186V to 225V), indicating the linear region where the core is not yet saturated.
- **Above 2.3A:** The rate of voltage increase slows down. For example, increasing current from 2.9A to 3.6A (a 0.7A increase) only raised the voltage by about 22V. This indicates that the machine is entering **Magnetic Saturation**. The domains in the iron core are fully aligned, and adding more current yields diminishing returns in flux generation.

7. Results

During this experiment, we gradually increased the excitation current of a three-phase synchronous generator, while keeping the rotor speed close to 1500 rpm. At each step, we recorded the terminal voltage.

Initially, the voltage rose quickly with small increases in excitation current—this reflects the linear part of the magnetization curve. However, once the excitation current passed around 2.3 A, the voltage gains became smaller. This suggests the magnetic core was entering saturation. In this region, even large increases in current caused only minor changes in voltage.

At the highest excitation level, we noticed a slight drop in rotor speed, likely due to the added magnetic strain on the system.

8. Conclusion

This experiment clearly showed how a three-phase synchronous generator behaves under no-load conditions. As expected, the terminal voltage increased with the excitation current, and magnetic saturation became evident at higher current levels.

The Open Circuit Characteristic (OCC) curve matched theory, revealing both the initial linear region and the saturation region, confirming the typical behavior of synchronous machines.

9. QUESTIONS

- **NO LOAD OPERATION ANSWER THE QUESTIONS**

QUESTION 1:

In a synchronous generator, the frequency of the induced voltage is directly related to the rotor speed according to:

$$f = \frac{P \cdot n}{120}$$

where
f is the frequency,
P is the number of poles,
nnn is the rotor speed in rpm.

During the no-load test, the speed is kept constant at its rated value (approximately 1500 rpm) to ensure a constant frequency of 50 Hz. This allows the effect of excitation current on the terminal voltage to be examined independently.

If the speed varies, both the frequency and the induced voltage change, making it impossible to correctly obtain the open circuit characteristic (OCC) curve. Therefore, constant speed operation is essential for accurate analysis.

QUESTION 2:

The induced voltage of a synchronous generator is proportional to the magnetic flux, which in turn depends on the excitation current:

$$\mathbf{E} \propto \mathbf{\Phi}$$

At low excitation currents, the magnetic core is unsaturated and the flux increases almost linearly with the excitation current. As a result, the terminal voltage rises rapidly.

However, beyond a certain excitation level, the iron core enters magnetic saturation. In this region, a large increase in excitation current produces only a small increase in magnetic flux. Consequently, the terminal voltage no longer increases linearly and tends to flatten out.

This behavior was clearly observed in the experiment, where increasing the excitation current above approximately 2.3 A resulted in diminishing voltage increments.

QUESTION 3:

When the excitation current is zero, there is no externally applied magnetic field in the rotor. However, due to residual magnetism in the iron core, a small voltage may still be induced at the stator terminals even at rated speed.

Although this residual voltage is usually low, it is unpredictable and may increase under certain conditions. Therefore, touching the generator terminals is dangerous and should always be avoided to prevent the risk of electric shock.

QUESTION 4:

Yes, a voltage is still observed at the generator terminals when the direction of the excitation current is reversed. Reversing the excitation current reverses the polarity of the magnetic field in the rotor, but the magnitude of the magnetic flux remains unchanged.

Since the induced voltage depends on the magnitude of the magnetic flux and the rotor speed, the output voltage magnitude remains approximately the same. Only the phase sequence or polarity of the generated voltage may change.

QUESTION 5:

The no-load operation curve, also known as the Open Circuit Characteristic (OCC), is obtained by plotting the terminal voltage against the excitation current while keeping the speed constant.

From the experimental data:

- At low excitation currents, the OCC curve is nearly linear, representing the unsaturated region of the magnetic core (air-gap line).

- As the excitation current increases, the curve bends and enters the saturation region.

- In the saturation region, further increases in excitation current result in only small increases in terminal voltage.

This curve confirms the magnetic saturation behavior of the synchronous generator and matches the theoretical OCC characteristic.

QUESTION 6:

In this experiment, the no-load operation of a three-phase synchronous generator was analyzed. It was observed that the terminal voltage increases with excitation current and that magnetic saturation occurs at higher excitation levels.

Maintaining a constant speed was essential for obtaining accurate results. The experimentally obtained open circuit characteristic curve showed good agreement with theoretical expectations, confirming the fundamental operating principles of synchronous generators under no-load conditions.

- **LOAD OPERATION ANSWER THE QUESTIONS**

QUESTION 1:
The terminal voltage of a synchronous generator changes depending on the type of load connected. When a resistive load is applied, the terminal voltage decreases moderately due to internal impedance effects.

With an inductive load, the terminal voltage decreases more significantly because the armature current produces a demagnetizing magnetic field that weakens the main flux.

In contrast, when a capacitive load is connected, the terminal voltage tends to increase. This is because the capacitive current produces a magnetizing effect, which strengthens the air-gap flux and raises the terminal voltage.

QUESTION 2:

The open-circuit terminal voltage Ea is defined as the voltage measured at the generator terminals under no-load conditions. In this experiment, the open-circuit voltage was obtained from the no-load operation test.

When the generator is loaded, the terminal voltage decreases or increases depending on the load type. The voltage drop is defined as the difference between the open-circuit voltage and the terminal voltage under load.

Voltage regulation is calculated using the following equation:

$$*\%Rg = \frac{Ea - Uf}{Uf}$$

%Rg: Voltage regulation percentage
Ea : Open circuit alternator voltage
Uf : Alternator terminal voltage at full load

where:

- Ea is the open-circuit alternator voltage,
- Uf is the alternator terminal voltage at full load.

In this experiment, the terminal voltage under load (Uf) was not directly recorded; therefore, numerical voltage regulation values could not be calculated. However, based on theoretical considerations and experimental observations:

- Voltage regulation is positive for resistive and inductive loads due to voltage drop.
- Voltage regulation is highest for inductive loads because of demagnetizing armature reaction.
- Voltage regulation may become negative for capacitive loads due to voltage rise.

QUESTION 3:

Armature reaction is the effect of the magnetic field produced by the stator (armature) current on the main magnetic field of the rotor. This interaction changes the air-gap flux and affects the terminal voltage of the alternator.

For resistive loads, armature reaction is mainly cross-magnetizing and causes a moderate voltage drop. For inductive loads, it is demagnetizing and significantly reduces the terminal voltage. For capacitive loads, armature reaction becomes magnetizing, which increases the main flux and raises the terminal voltage.

QUESTION 4:

To maintain a constant terminal voltage under different load conditions, the excitation current of the synchronous generator can be adjusted. Increasing the excitation current compensates for voltage drops caused by inductive loads, while decreasing excitation current prevents excessive voltage rise under capacitive loading.

In practical power systems, this regulation is achieved automatically using an Automatic Voltage Regulator (AVR), which continuously controls the excitation current.

QUESTION 5:

During loaded operation, the synchronous generator supplies active and reactive power to the connected load. As the load increases, the stator current increases, leading to higher internal voltage drops and stronger armature reaction effects.

Experimental observations showed that reactive power was dominant over active power, indicating an inductive load characteristic. The calculated power factor values were very low, confirming that the generator operated under strongly inductive conditions. These characteristics are consistent with the theoretical behavior of synchronous generators under load.

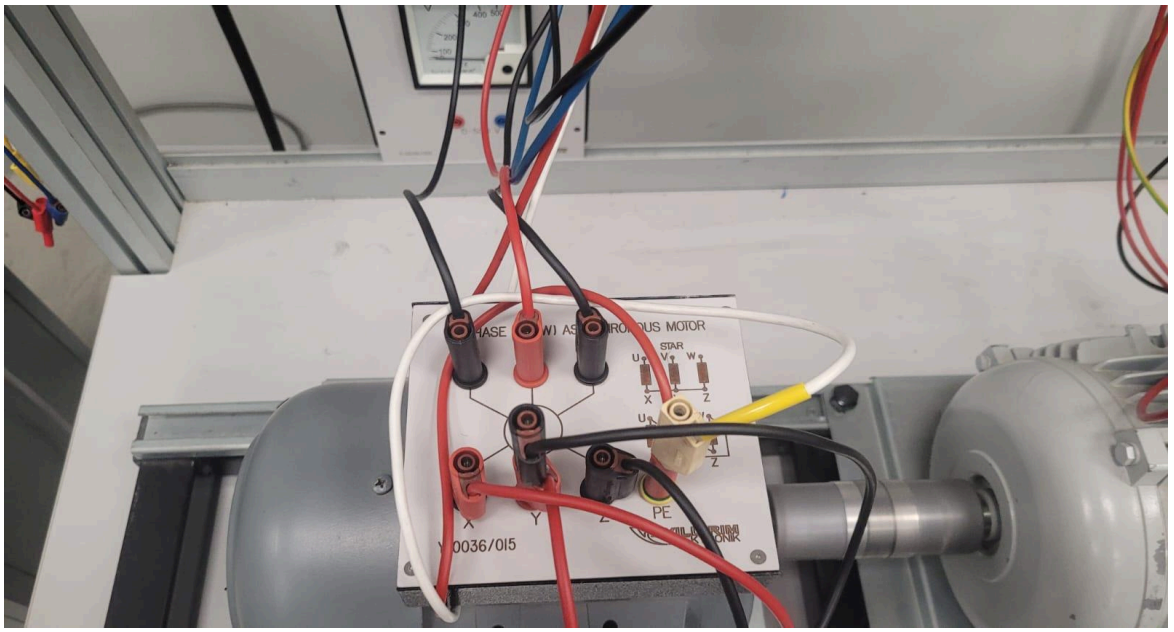
QUESTION 6:

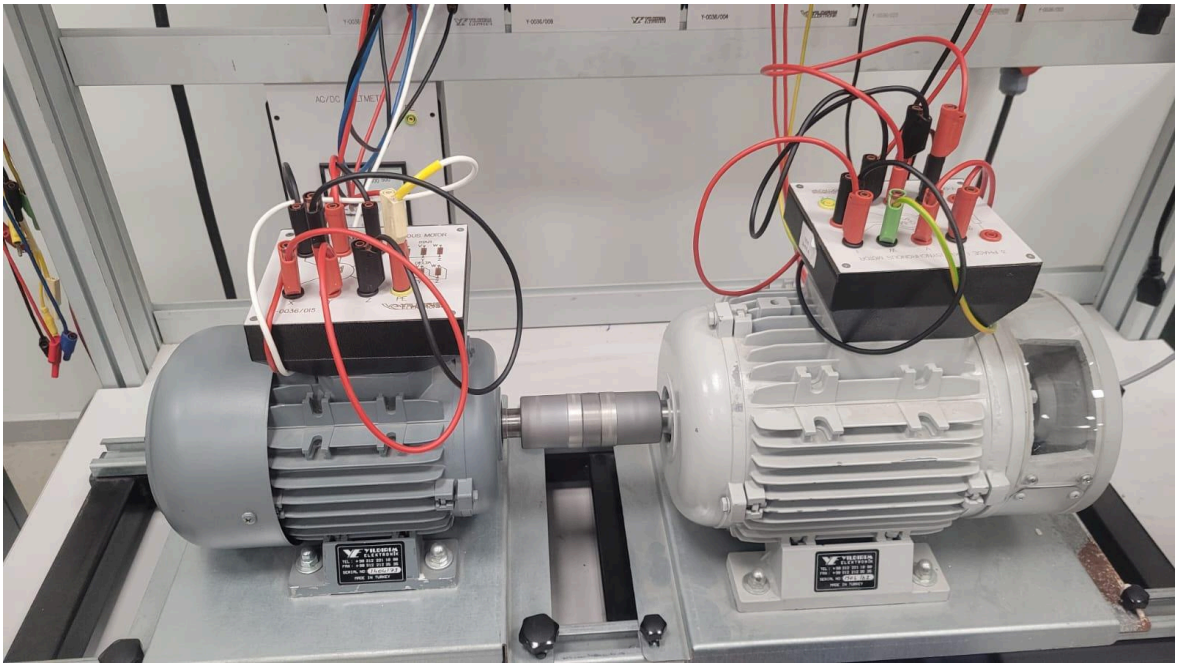
In this experiment, the loaded operation characteristics of a three-phase synchronous generator were analyzed. It was observed that the generator terminal voltage behavior strongly depends on the load type and excitation level.

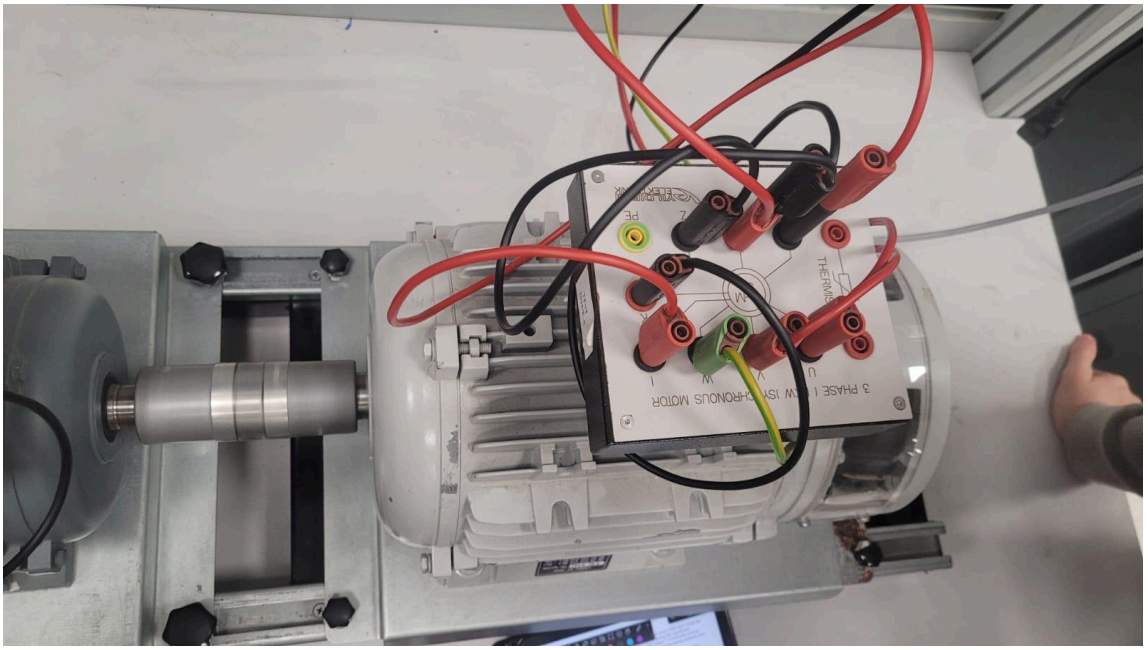
Inductive loading resulted in significant reactive power and voltage reduction due to demagnetizing armature reaction. The experimental observations were consistent with theoretical expectations, highlighting the importance of excitation control and voltage regulation in synchronous generator operation.

10. EXPERIMENT PHOTOS

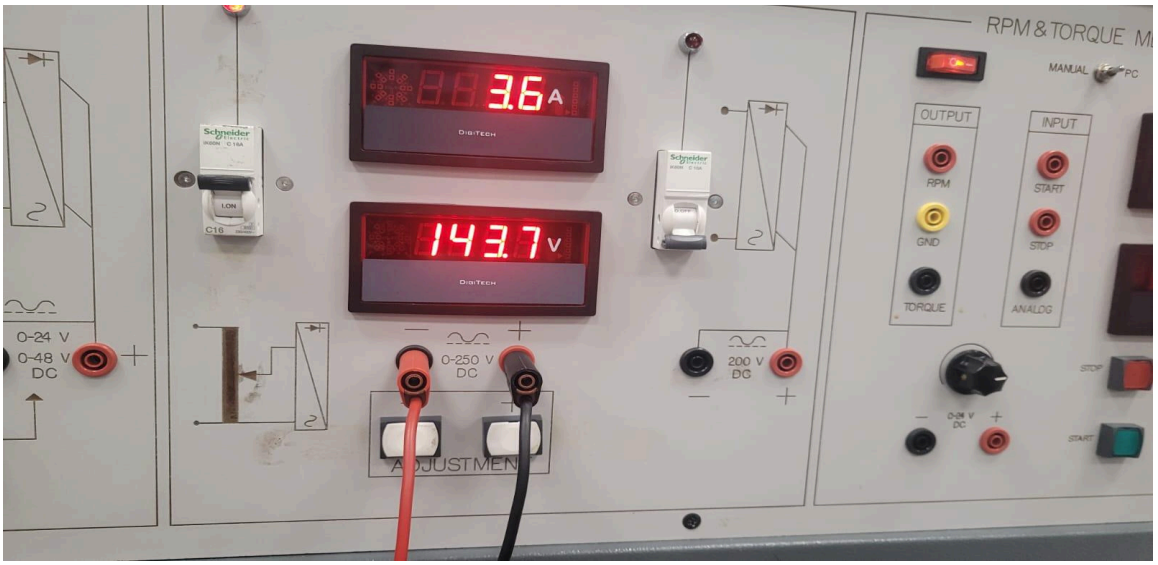
EXPERIMENT DESIGN







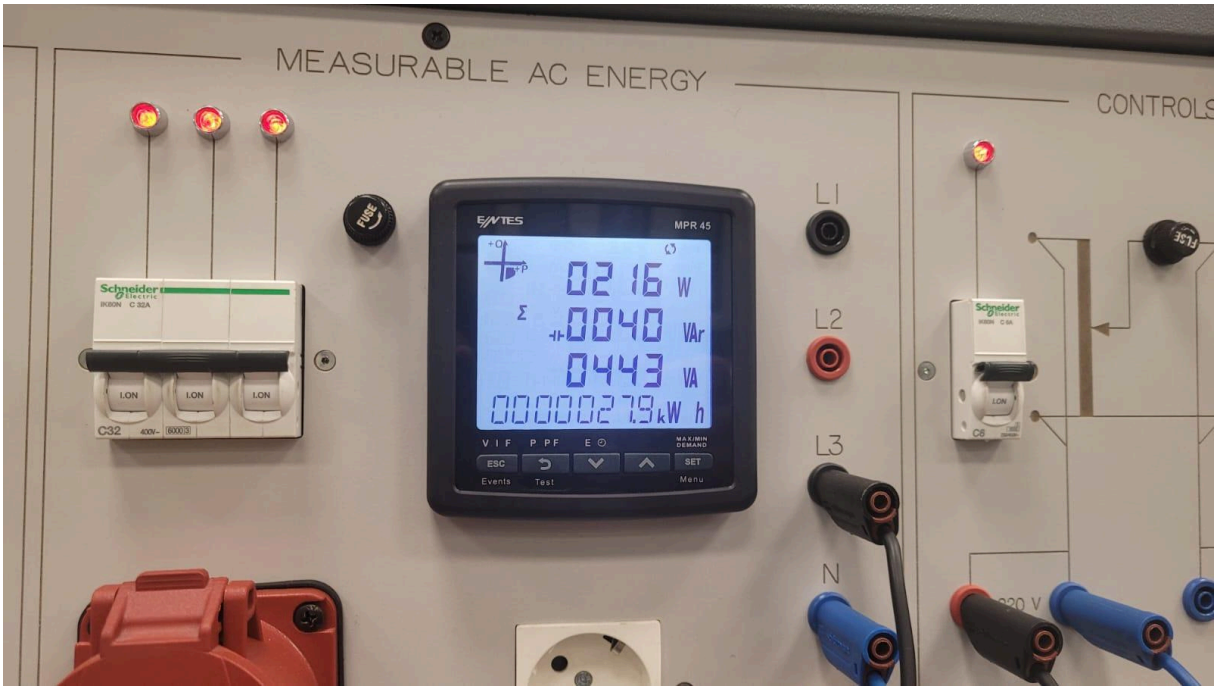
NO-LOAD OPERATION

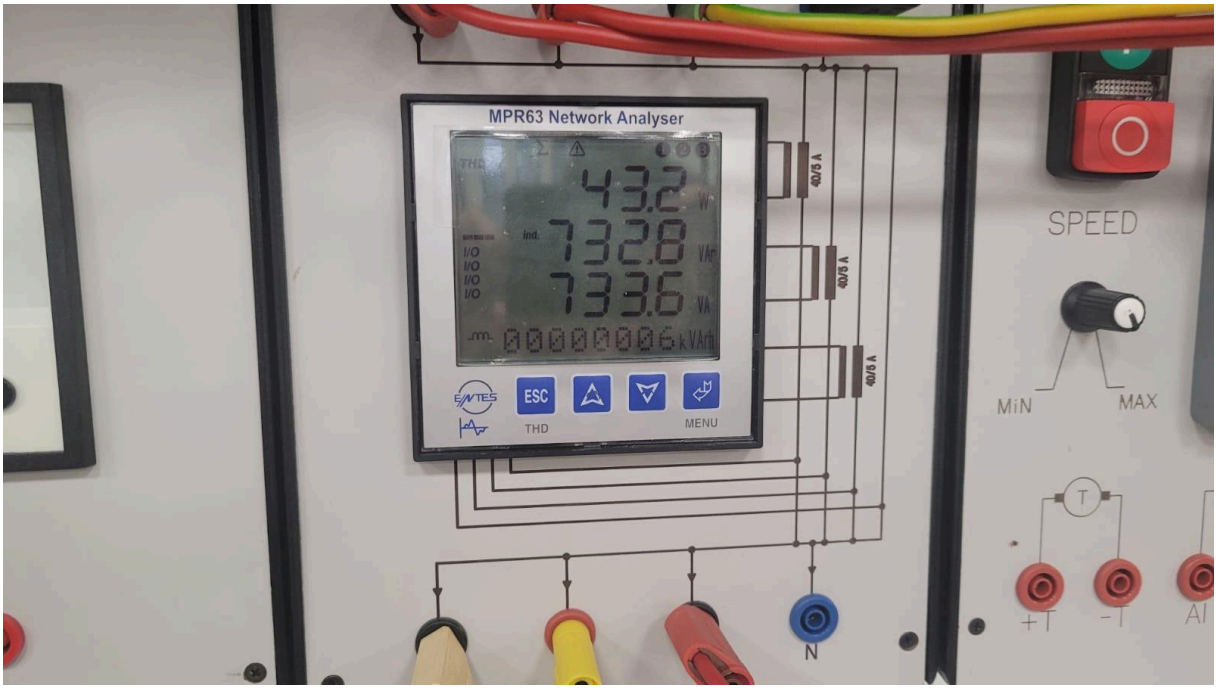






LOAD OPERATION


MEASURABLE AC ENERGY
CONTROLS
EMTES
MPR 45
0216 W
0040 VAr
0443 VA
00000279 kW h
VIF PPF E
MAX/MIN DEMAND
ESC
SET
Events
Test
Menu
L1
L2
L3
N
Schneider Electric
I.ON
C32
FUSE


MPR63 Network Analyser
43.2 W
732.8 VAr
733.6 VA
00000006 kVArh
ESC
THD
MENU
SPEED
MiN
MAX
N
+T
-T

References

[1] https://en.wikipedia.org/wiki/Synchronous_impedance_curve